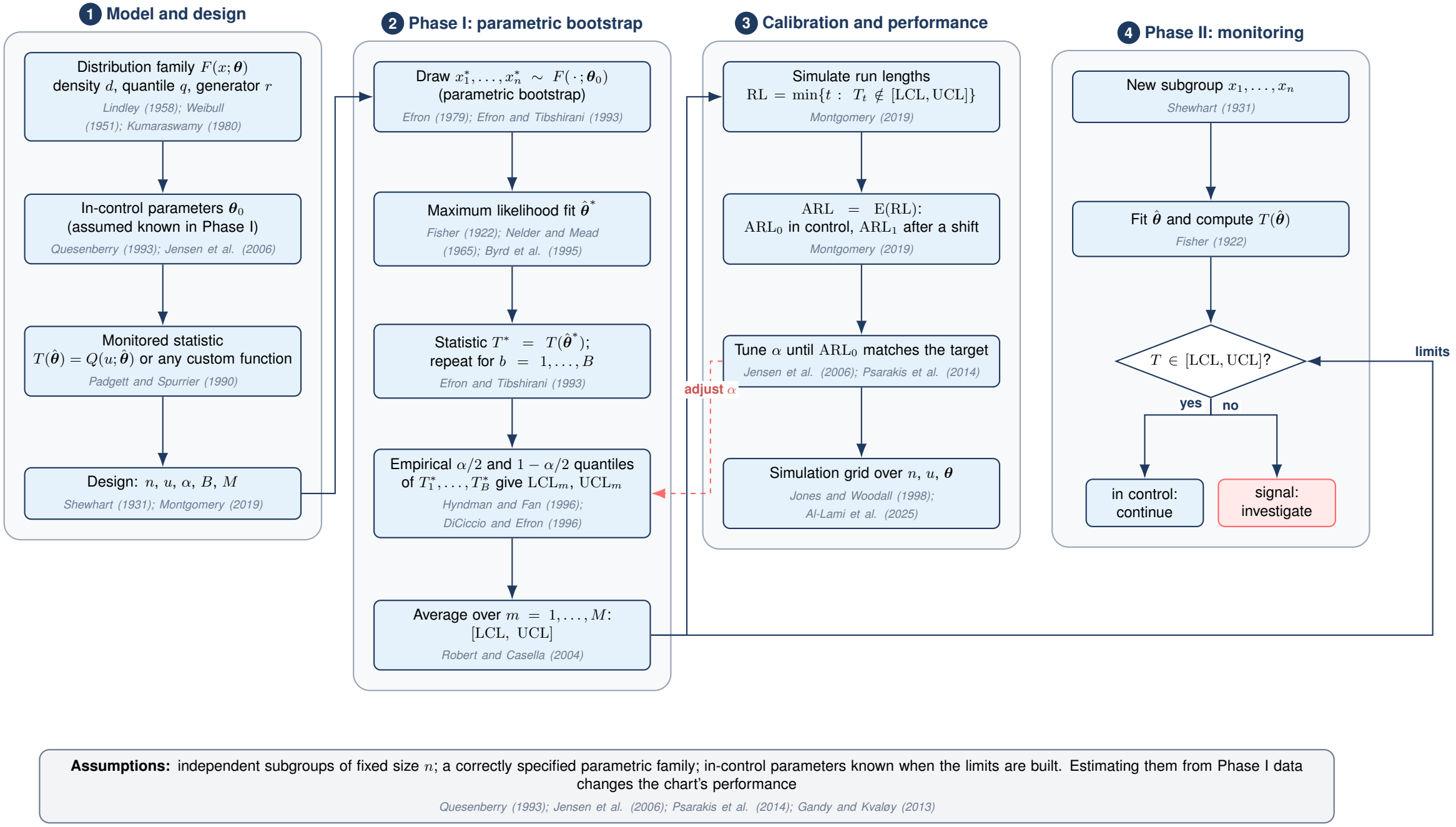

1 Model and design
Distribution family $F(x;\boldsymbol{\theta})$
density d, quantile q, generator r
Lindley (1958); Weibull (1951); Kumaraswamy (1980)
In-control parameters $\boldsymbol{\theta}_0$
(assumed known in Phase I)
Quesenberry (1993); Jensen et al. (2006)
Monitored statistic
$T(\hat{\boldsymbol{\theta}}) = Q(u;\hat{\boldsymbol{\theta}})$ or any custom function
Padgett and Spurrier (1990)
Design: n, u, α, B, M
Shewhart (1931); Montgomery (2019)
2 Phase I: parametric bootstrap
Draw $x_1^*, \ldots, x_n^* \sim F(\cdot;\boldsymbol{\theta}_0)$
(parametric bootstrap)
Efron (1979); Efron and Tibshirani (1993)
Maximum likelihood fit $\hat{\boldsymbol{\theta}}^*$
Fisher (1922); Nelder and Mead (1965); Byrd et al. (1995)
Statistic $T^* = T(\hat{\boldsymbol{\theta}}^*)$;
repeat for $b = 1, \ldots, B$
Efron and Tibshirani (1993)
Empirical $\alpha/2$ and $1-\alpha/2$ quantiles
of $T_1^*, \ldots, T_B^*$ give LCL_m, UCL_m
Hyndman and Fan (1996); DiCiccio and Efron (1996)
Average over $m = 1, \ldots, M$:
[LCL, UCL]
Robert and Casella (2004)
3 Calibration and performance
Simulate run lengths
$\mathrm{RL} = \min\{t : T_t \notin [\mathrm{LCL}, \mathrm{UCL}]\}$
Montgomery (2019)
ARL = E(RL):
ARL_0 in control, ARL_1 after a shift
Montgomery (2019)
Tune α until ARL_0 matches the target
Jensen et al. (2006); Psarakis et al. (2014)
adjust α
Simulation grid over n, u, $\boldsymbol{\theta}$
Jones and Woodall (1998); Al-Lami et al. (2025)
4 Phase II: monitoring
New subgroup $x_1, \ldots, x_n$
Shewhart (1931)
Fit $\hat{\boldsymbol{\theta}}$ and compute $T(\hat{\boldsymbol{\theta}})$
Fisher (1922)
$T \in [\mathrm{LCL}, \mathrm{UCL}]$?
yes
no
in control: continue
signal: investigate
limits
Assumptions: independent subgroups of fixed size n; a correctly specified parametric family; in-control parameters known when the limits are built. Estimating them from Phase I data changes the chart's performance
Quesenberry (1993); Jensen et al. (2006); Psarakis et al. (2014); Gandy and Kvaløy (2013)